

Shuai Dong

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SUMMARY

Software Engineer & Ph.D. candidate (CS) with 5+ years building real-time communication and automation on Linux. Hands-on with TCP/UDP socket programming, concurrent pipelines, and reliability on low-bandwidth/high-latency links. Designed Python/C++ tooling for large-scale tests, stress/performance validation, and metrics. Integrated ROS/MOOS-IvP with custom network stacks for remote monitoring/control of multi-robot (AUV) systems. Familiar with streaming/teleop concepts, Wi-Fi/cellular connectivity, and telemetry dashboards.

EDUCATION

The University of Alabama, Tuscaloosa, AL
Ph. D. of Computer Science (GPA: 3.9)

August 2019 – Present

The University of South Alabama, Mobile, AL
Master of Computer Science (GPA 3.8) (Visiting Scholar)

July 2018 – July 2019

TECHNICAL SKILLS

- **Skills:** Python (automation, scripting, data analysis), C, C++, Java | Bash/CLI
- **Test & Automation:** Pytest, automated regression testing, performance/stress testing, CI/CD concepts
- **Networking & Protocols:** TCP/UDP socket programming, routing/MAC protocol design, IOT/telematics simulations, REST APIs, MQTT (familiar)
- **Systems:** Linux (CLI, benchmarking, debugging, shell scripting), virtualization, ROS, MOOS-IvP
- **Tools:** Wireshark, tcpdump, iperf, AquaNet, NS-3, Git
- **Other:** Optimization algorithms, system monitoring, cross-functional collaboration

PROFESSIONAL EXPERIENCE

The University of Alabama, Tuscaloosa, AL

August 2019 – Present

(1) Automated Test & System Reliability

- a. Designed Python-based automation scripts to run and monitor large-scale network simulations, integrating logging, benchmarking, and performance validation.
- b. Developed regression-style test workflows to reproduce experimental results, ensuring reproducibility and early detection of errors.
- c. Automated experiment pipelines with Python and shell scripting, analogous to CI/CD workflows.

(2) Protocol & Connectivity Testing:

- a. Implemented routing and adaptive power control protocols in C/C++ within Linux environments; tested under stress conditions to evaluate reliability and efficiency.
- b. Used Linux networking tools (tcpdump, Wireshark) to debug communication issues, analyze packet loss, and optimize throughput.
- c. Built hybrid testbeds (real + emulated) to evaluate protocol scalability and integrated them with middleware (ROS, MOOS-IvP) for IoT-style robotic communication.

(3) Performance & Optimization:

- a. Conducted automated large-scale stress and performance tests to measure system efficiency, achieving ~40% increase in network lifetime and ~30% reduction in per-node energy waste.
- b. Applied optimization algorithms and data analysis pipelines (Python + NumPy/Pandas) to interpret results and refine system parameters.

SELECTED PROJECTS

(1) Library Management System

- a. Using HTML/CSS design the frontend and set corresponding functional buttons.
- b. Built a full-stack CRUD application with Java servlets and MySQL backend, optimized queries for performance

(2) NSF Project for Underwater Robotic Testbed:

- a. Built a C/C++ driver/daemon bridging ROS/MOOS-IvP and the network stack, converting pub/sub topics into acoustic frames with packetization/framing, checksums, thread-safe queues, rate control, and retransmission for low-bandwidth, high-latency links.
- b. Automated multi-node experiments with Python/bash; added structured logging, health metrics, and topic→port mapping to ensure reliable multi-AUV data exchange over AquaNet-emulated channels.
- c. Integrated ROS/MOOS-IvP with the comms stack to enable end-to-end protocol evaluation and regression-style testing in controlled, scalable environments.